Interactive Systems

Los Del DGIIM, `losdeldgiim.github.io`

Doble Grado en Ingeniería Informática y Matemáticas
Universidad de Granada

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Arturo Olivares Martos

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Índice general

1. Research Methods for Interactive Systems Design

When designing interactive systems, it is crucial to understand the design space:

- Users: needs and requirements.
- Situation: what is the current state of the problem.
- Context: location, legislation, culture, etc.

There are different types of research methods:

- Quantitative / Qualitative: Quantitative methods involve collecting and analyzing numerical data, while qualitative methods focus on understanding human behavior and the reasons behind it.
- In Situ / In Laboratory: In situ research is conducted in the natural environment of the users, while laboratory research takes place in a controlled setting.
- Unstructured / Structured: Unstructured research allows for more flexibility and exploration, while structured research follows a predefined plan and methodology.

In order to design effective interactive systems, *triangulation* is often used, which involves combining multiple research methods to gain a comprehensive understanding of the design space. Using these research methods, theory and practice can be combined:

- Induction: Theory is derived from practice, and it is used to generate new insights and understanding about the design space.
- Deduction: Theory is used to inform practice and create hypotheses, and it is applied to guide the design and development of interactive systems.

During the following sections, we will explore various research methods that can be employed to gather all the needed information.

1.1. Ethnography

Ethnography is a research method that is founded on participation rather than observation, with the goal of not only understanding what people do but also why they do it and how they feel about it. It involves immersing oneself in the users' environment to gain insights into their behaviors, motivations, and experiences.

- It is a qualitative, in situ and unstructured research method that uses induction.

There are two main types of ethnography:

Observing Participation The researcher observes the users' activities and interactions without actively engaging in them. This allows for an objective perspective on the users' behaviors and experiences.

Participating Observation The researcher is actively involved in the users' activities and interactions, allowing for a deeper understanding of their experiences and perspectives. This method has the risk of going "native", where the researcher becomes too immersed in the users' environment and loses objectivity.

Some methods used in ethnography include:

- Field Diary: Personal notes taken by the researcher during their immersion in the field, capturing observations, thoughts, and reflections.
- Cameras: Used to document the environment and interactions, providing visual data for analysis. One important disadvantage is the *Hawthorne Effect*, which refers to the phenomenon where individuals modify their behavior in response to being observed, potentially leading to biased data.
- Diary Studies: Participants keep a diary of their activities and experiences.
- People receive package with material for documentation: Participants are given materials to document their experiences, such as cameras or notebooks. After a certain period, the researcher collects the materials and analyzes the data to gain insights into the users' behaviors and experiences.

1.2. Action Research

AR is a research method that is based on the idea of taking action to solve a problem while simultaneously conducting research to understand the problem and its context. As opposed to ethnography (where the researcher is an insider but does not change the environment), in AR the researcher is an insider and changes the environment.

- It is a qualitative, in situ and unstructured research method. It uses induction, but its generalization is difficult because it is context-specific.

There are three main types of AR:

Technical AR The researcher intervenes based on his technical expertise, sharing his knowledge with the users to solve a specific problem.

Practical-deliverative AR The researcher tries to understand the users' needs and requirements, and then tries to solve the problem based on that understanding (not necessarily based on his previous technical expertise).

Critical-emancipatory AR The researcher tries to understand the users' needs and requirements, and then tries to empower the users to solve the problem by themselves, without necessarily providing a solution directly.

AR involves a cyclical process of planning, acting and examining the results, with feedback loops that allow for continuous improvement and adaptation of the interventions. The three main steps of AR are:

1. Planning: The researcher identifies a problem and develops a plan to address it. *"Unfreezing"* the current state of the problem is crucial to allow for change, as it involves breaking down existing habits and behaviors that may be resistant to change.

2. Acting: The researcher implements the plan and takes action to address the problem. This may involve introducing new technologies, processes, or interventions to improve the situation.

3. Examining the results: The researcher evaluates the outcomes of the actions taken and reflects on the effectiveness of the interventions. *"Refreezing"* the new state of the problem is important to ensure that the changes are sustained over time, as it involves reinforcing new habits and behaviors that have been established.

1.3. Research through Design

RtD is a research method that is based on the idea of using design as a means of inquiry and exploration. It involves creating prototypes and artifacts not to solve a specific problem, but to gain insights and understanding about the design space and the users' needs and requirements.

- It is a qualitative, in situ and usually unstructured research method.

It is a cyclical process that involves four main steps:

1. Selecting a problem: The researcher identifies a problem or area of interest to explore through design.

2. Designing a solution: The researcher creates a prototype or artifact that addresses the selected problem or area of interest.

3. Evaluating the solution: The researcher evaluates the prototype or artifact to gain insights into the design space and the users' needs and requirements.

4. Reflecting on the results: The researcher reflects on the insights gained from the evaluation and uses them to inform the next iteration of the design process.

It has several benefits, such as:

- It shows technical opportunities and challenges that engineers may not have thought of.

- Exposes the gaps between the theory and practice of design.
- Changes current practices, therefore opening up new design spaces.
- Reveals designer patterns and assumptions, which can be used to inform future design decisions.

1.4. Field Deployment

Field deployment is a research method that involves deploying a prototype or artifact in the users' natural environment to gain insights into its use and effectiveness. It allows researchers to observe how users interact with the prototype in real-world settings, providing valuable feedback for further design iterations. Testing it in the lab is not enough, as it may not capture the complexities and nuances of real-world use.

- It is a qualitative, in situ and usually structured research method.

There are 7 main steps in field deployment:

1. Finding a Setting: There are three different types of settings:
   - *Convenience*: Deployment within the researcher's own environment. Not representative, and participants may be biased.
   - *Semi-Controlled*: It involves knwon and unknown participants. It can be more representative, but it may still have some biases.
   - *In the Wild*: Deployment in the users' natural environment to almost unknown participants. It is the most representative, but needs more robust prototypes.
2. Defining Goals: The fundamental goal is increasing understanding of the prototype that generalizes beyond the specific deployment.
3. Recruiting Participants: They should be representative of the target user population, also taking into account the researcher's group.
4. Designing Data Collection Instruments: Both qualitative (e.g., interviews, observations) and quantitative (e.g., measured time to complete a task, error rates) data collection instruments should be designed to gather comprehensive insights into the users' interactions with the prototype.
5. Conducting the Deployment: Pilot tests should be conducted to identify and address any issues with the prototype or data collection instruments before the full deployment.
6. Ending the Deployment: When the deployment is over, the product is usually taken away from the users in order to reduce the cost of mantaining its servers. In addition, they are usually not allowed to be bought by the users, as usually researchers do not want to sell products but only to gain insights.

Taking the prototype away from the users can be a difficult process, as it may have become an integral part of their daily lives. It is important to plan for this transition and provide support to users as they adjust to the change.

7. Analyzing the Data

1.5. Experimental Research

Experimental research is a research method whose aim is to know the cause-effect relationship between two or more variables. It involves manipulating one variable (called *independent variable*) and observing the effect on one or more other variables (called *dependent variables*), while controlling for other factors that may influence the results. In order to conduct a good experiment, a prediction (called *hypothesis*) should be made about the expected relationship between the variables.

- It is a quantitative, in laboratory and structured research method.

Two different factors should be taken into account when designing an experiment:

- Validity: It refers to the extent to which the results of an experiment can be generalized to other settings, populations, and times. There are several types of validity, as population validity, ecological validity, temporal validity, etc.
- Reliability: It refers to the consistency and stability of the results of an experiment. An experiment is considered reliable if it produces the same results when repeated under the same conditions.

As explained before, hypotheses are predictions about the expected relationship between variables, and they are essential for designing and conducting experiments. They should be *precise*, *meaningful* (based on theory), *testable*, and *falsifiable*. In order to test a hypothesis, the null hypothesis (which states that there is no relationship between the variables) should be rejected.

There are two different approaches to research in experimental research:

- *Between-Subject Design*: Different groups of participants are exposed to different conditions of the independent variable. This design allows for comparisons between groups, but it may require a larger sample size to achieve statistical power.
- *Within-Subject Design*: The same group of participants is exposed to all conditions of the independent variable. They are firstly exposed to one condition, and then to the other condition. This design allows for comparisons within the same group, but it may be affected by order effects (e.g., fatigue, learning).

  In order to mitigate order effects in within-subject designs, counterbalancing can be used, which involves varying the order of conditions across participants to ensure that any potential order effects are evenly distributed and do not bias the results. There are two main types of counterbalancing:

  - *Complete Counterbalancing*: All possible orders of conditions are used, which can be impractical for a large number of conditions, as $x!$ orders are needed (where x is the number of conditions).

- *Latin Square Counterbalancing*: A more efficient method that uses a Latin square design to ensure that each condition appears in each position an equal number of times, reducing the number of orders needed to x (where x is the number of conditions).

1.6. Survey Research

Survey research is a research method that involves collecting data from a sample of individuals using standardized questionnaires or interviews.

- It is a quantitative, both in situ and in laboratory, and a structured research method.

Surveys are used to gather information about people's attitudes, beliefs, behaviors, and characteristics... but are not good when understanding the reasons behind those attitudes, beliefs, behaviors, and characteristics. Therefore, they are often used in combination with other research methods (e.g., ethnography, action research) to gain a more comprehensive understanding of the design space.

- Surveys are used to decide if the results of the research methods can be generalized to a larger population.
- The other research methods are used to understand the reasons or trends behind the attitudes, beliefs, behaviors, and characteristics of the users.

There are different stages in survey research:

1. Research Goals and Constructs: Firstly, the research goals should be defined, which will guide the entire survey research process. Then, the constructs that will be measured should be identified.
2. Population and Sampling: There are different scopes of people that can be surveyed. The population groups considered, from the most general to the most specific, are:
   - *Population*: The entire group of individuals that the researcher is interested in studying.
   - *Sampling Frame*: The individuals that the researcher has access to and can potentially include in the survey.
   - *Sample*: The individuals that are actually asked to participate in the survey. Not all individuals in the sampling frame may be included in the sample, as some may be excluded due to various reasons (e.g., ethical reasons, ineligibility). There are different sampling methods that can be used to select the sample:
     - *Probability Sampling*: Each individual in the sampling frame has the same probability of being selected for the sample. This method allows for generalization of the results to the population, but it may be more time-consuming and expensive.

- *Non-Probability Sampling*: Individuals are selected based on non-random criteria, such as convenience or judgment. This method is less time-consuming and expensive, but it may introduce bias and limit the generalizability of the results.

- *Respondents*: The individuals that actually respond to the survey. The response rate (i.e., the percentage of respondents out of the total sample) is an important factor to consider, as a low response rate can lead to non-response bias and affect the generalizability of the results.

3. <u>Questionnaire Design and Bias</u>:

   There are two main types of questions that can be used in a survey:

   - *Close Ended Questions*: These questions provide respondents with a set of predefined response options to choose from. They are easier to analyze and easier for respondents to answer, but they may limit the depth and richness of the data collected. This should be the most used type of question in surveys. They can be further categorized into:
     - Single-choice questions.
     - Multiple-choice questions.
     - Likert scale questions, aka rating questions.
     - Ranking questions.
   - *Open Ended Questions*: These questions allow respondents to provide their own answers in their own words. They can provide richer and more detailed data, but they may be more difficult to analyze and may require more time for respondents to answer.

   When designing a questionnaire, it is important to be aware of potential biases that can affect the validity of the results. A bias is a systematic error that can lead to inaccurate or misleading results. Some common types of bias in survey research include:

   - *Satisficing Bias*: Respondents may use a low cognitive effort strategy to answer survey questions, such as selecting the first acceptable response option.
   - *Acquiescence Bias*: Respondents may have a tendency to agree with statements or questions, regardless of their true feelings or opinions. Questions with `agree/disagree` response options are particularly susceptible to this bias.
   - *Social Desirability Bias*: Respondents may provide answers that they believe are socially acceptable or desirable, rather than their true thoughts or feelings.
   - *Response Order Bias*: The order in which response options are presented can influence respondents' choices, with a tendency to select options that are presented first or last.

- *Question Order Bias*: The order of questions can influence respondents' answers, as earlier questions may prime certain thoughts or feelings that affect responses to later questions.

In addition, when designing a questionnaire, there are already existing validated questionnaires that can be used to measure certain constructs. Using these existing questionnaires can save time and effort, and can also improve the validity and reliability of the results, as these questionnaires have already been tested and validated in previous research.

4. Review and Survey Pretesting:

   There are two main types of pretesting that can be conducted to evaluate the questionnaire before it is administered to the full sample:

   - *Cognitive Pretesting*: This involves asking a small group of respondents to complete the questionnaire while thinking aloud, allowing the researcher to identify any issues with question wording, comprehension, or response options.
   - *Pilot Testing*: This involves administering the questionnaire to a small sample of respondents that are similar to the target population, allowing the researcher to identify any issues with the survey administration process, such as the length of time it takes to complete the survey or any technical issues with the survey platform.

5. Implementation and Launch:

   There are different well known platforms that can be used to implement and launch a survey, such as Qualtrics, SurveyMonkey, Google Forms, etc.

6. Data Analysis and Reporting:

   When analyzing survey data, it is important to consider the type of data collected (e.g., nominal, ordinal, interval, ratio) and to use appropriate statistical methods for analysis. Some additional metrics can be analyzed, such as:

   - *Data Logging*: recording additional information about respondents' interactions with the survey, such as the time taken to complete the survey or the order of responses. This can provide valuable insights into respondents' behavior and can help identify potential issues with the survey design.
   - *Response Rates*: Analyzing response rates can help identify potential biases in the sample and can inform strategies for improving response rates in future surveys.

2. Human Robot Interaction

During this chapter, we will be talking about the interaction between humans and robots, and how to design interfaces that allow for effective communication and collaboration between the two. However, before we dive into it, it is important to understand what a *robot* is. A robot is a multi-purpose machine that can be programmed to perform a variety of tasks emulating living beings. They are therefore mobile and autonomous. The three esential aspects of a robot are:

- Sensing: The ability to perceive the environment through sensors, such as cameras, microphones, and touch sensors.
- Thinking: The ability to process information and make decisions based on the data received from the sensors.
- Acting: The ability to perform actions based on the decisions made by the robot, such as moving, manipulating objects, or communicating with humans.

Since robots are becoming more common in our daily lives, it is crucial to correctly understand the Human Robot Interaction (HRI), specially because of the *social* aspect of it. Even though robots may have a correct physical interaction because of their sensors and actuators, they may fail to interact with humans in a way that is socially acceptable or effective. HRI is then a multidisciplinary field that focuses on developping robots that can interact with humans in a natural and intuitive way.

2.1. Classifications in HRI

When working in HRI, plenty of different classifications should be taken into account. However, the most important ones are:

- Robot classification depending on their interaction capabilities.
- Human classification depending on their role in the interaction.
- Interaction classification depending on the type of interaction between humans and robots.

Depending on their interaction capabilities, we can classify robots into different categories:

- Bounded Autonomy: Robots with limited autonomy that operate within predefined constraints.

- Teleoperation: Robots controlled remotely by a human operator.
- Supervised Autonomy: Robots that operate autonomously but under human supervision.
- Adaptive Autonomy: Robots that can adapt their behavior based on the environment and user needs.
- Virtual Symbiosis: Robots that collaborate with humans in a way that enhances both parties' capabilities. The robot is a co-worker.

Depending on the role of humans in the interaction, we can classify them into different categories:

- Supervisor: The human is responsible for overseeing the robot's actions and ensuring that it operates safely and effectively.
- Operator: The human is responsible for controlling the robot's actions, either directly or through a user interface.
- Mechanic: The human is responsible for maintaining and repairing the robot, ensuring that it remains in good working condition.
- Peer: The human is a collaborator with the robot, working together to achieve a common goal.
- Bystander: The human avoids interaction with the robot, either because they are not interested or because they are afraid of it.

Depending on the type of interaction between humans and robots, we can classify them into different categories:

- Coexistence: The human and the robot share the same environment at the same time, but they do not interact with each other.
- Cooperation: The human and the robot share the same environment at the same time, and they share a common goal, but they do not contact or interact with each other to achieve it.
- Collaboration: The human and the robot share the same environment at the same time, share a common goal, and they interact with each other to achieve it.

2.2. Development of a Robot

When developing a robot, there are several different aspects that should be taken into account. In order to design it, a modular approach should be taken, and the *three main components of the robot* are considered from the sense-think-act approach:

- Perception (Hardware and Software):

  Both self monitoring and environment perception are crucial for a robot to interact effectively with humans.

  - *Self monitoring*: The robot should be able to monitor its own state, such as its battery level, motor status, and sensor readings.
  - *Environment perception*: The robot should be able to perceive the environment through sensors, such as cameras, microphones, and touch sensors.

- Cognition (Software):

  The robot should have a cognitive architecture that allows it to process information and make decisions based on the data received from the sensors. It runs on operating systems, such as ROS (Robot Operating System), that provide a framework for developing and deploying robot software.

  Their skills are usually preprogrammed, but they can also learn from experience and adapt to new situations (machine learning). Most of their learning capabilities are offline, but active research is being done in online learning, which allows robots to learn and adapt in real-time.

- Action (Hardware and Software):

  In order to perform actions, the robot should have actuators, such as motors and grippers, that allow it to move and manipulate objects.

Once the three main components of the robot are defined, the design of the interaction can be done. Two different approaches are:

- Inside-out approach: The design starts by building the technology, and the interaction is designed around it. It is later tested by social scientists, who evaluate the interaction and provide feedback to the engineers.
- Outside-in approach: The design starts by understanding the social context and the needs of the users, and then the technology is developped to meet those needs.

When designing the robot, there are different aspects that should be considered regaring the interaction with humans. Some of the most important ones are:

- Morphology: The physical design of the robot, which can affect how humans perceive and interact with it. It should take into account several factors:

  - The form and their function should be consistent.
  - Android robots (those that physically resemble humans) should be designed with care, as they may be antropomorphic. In addition, the *uncanny valley* phenomenon should be taken into account (Figure 2.1), which describes the discomfort that humans may feel when interacting with robots that are very close to human appearance, but not quite there.
  - Antropomorphic robots (those that are attributed with human traits or emotions) should be designed with care, as they may create unrealistic expectations about their capabilities. There are several questionnaires that can be used to evaluate the antropomorphism of a robot, such as the Godspeed questionnaire.

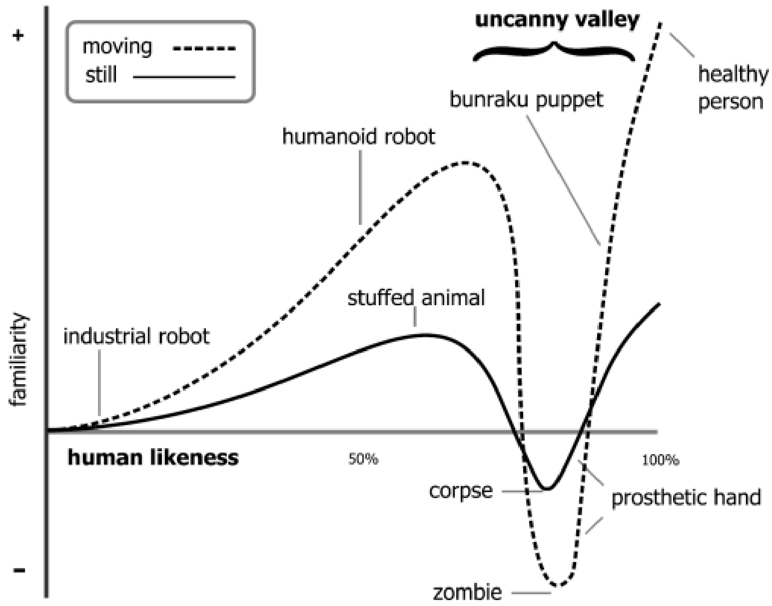


Figura 2.1: The uncanny valley phenomenon.

It should be noted that android are more tend to be antropomorphic, but these are not the same thing. For example, a robot that looks like a dog may be antropomorphic, but it is not an android. In addition, computers can also be antropomorphized when humans shout at them, but they are not robots.

- Affordances: The design of the robot should provide clear affordances that indicate how it can be interacted with. For example, a robot with a handle may indicate that it can be picked up, while a robot with a screen may indicate that it can be interacted with through touch.

  An important concept is the idea of *underpromise and overdeliver*, where the robot should be designed to exceed the expectations of the users. This can be achieved by providing additional functionality or features that are not explicitly stated in the design, but that enhance the user experience.

  Lastly, the *interaction should expand the functionality* of the robot beyond its initial design. For example, a robot designed for cleaning may also be able to provide companionship and social interaction.

- Human-centered design: The design of the robot should be centered around the needs and preferences of the users, using prototypes and user testing to iteratively improve the design.

  Robot design is usually a *wicked problem*, as there is no single solution that can satisfy all the requirements and constraints. Therefore, it is important to involve users in the design process to ensure that their needs and preferences are taken into account.

- Culture: The design of the robot should take into account the cultural context in which it will be used, as different cultures may have different expectations and preferences regarding robots.

2.3. Interaction Modalities

When designing the interaction between humans and robots, different modalities can be used to facilitate communication and collaboration. In the following sections, we will talk about the most common interaction modalities used in HRI.

2.3.1. Spatial Interaction

Placing a robot in people space should consider people's preferences. There are plenty of aspects that should be taken into account, such as:

- Localization and Navigation: Robots should be able to localize themselves in the environment, so they should be able to build a map of the environment trough sensors, such as cameras and lidar, and use that map to navigate and avoid obstacles. They should also be able to localize their different body parts, such as their arms and grippers, to perform tasks that require manipulation.

- Social Positioning: Navigation in the environment is another challenge, as the robot should be able to navigate in a way that is socially acceptable and does not cause discomfort to humans. Proxemics defines the different zones of personal space that humans have, and robots should be designed to respect those zones.

  In addition, phisically safe motion is not neccesarily perceived as safe by humans, so the robot should be designed to move in a way that is perceived as safe by humans.

- Initiating Interaction: Every interaction should be initiated. Although it may be trivial for humans, it is not for robots. For instance, frontal approach is more likely to initiate an interaction than a lateral approach.

- Intent Sharing: Robot motion trajectories should be designed to share the robot's intent with humans (even though they may not be the most efficient ones).

2.4. Nonverbal Interaction

Nonverbal communication is an essential aspect of human-robot interaction, as it can convey information and emotions that may not be easily expressed through verbal communication. The *Theory of Mind* is a crucial aspect of nonverbal interaction, as it is the ability to attribute mental states, such as beliefs, desires, and intentions, to oneself and others. Even though humans have a natural ability to understand the mental states of others (usually from a very young age), it is a challenge for robots to develop this ability.

Given this importance, robots should be designed to both express and understand nonverbal communication. Some of the most important types of nonverbal communication in HRI are:

- Gaze: Both humans' and robots' gaze can convey information about attention, interest, and intention. Robots should be designed to use their gaze and to interpret the gaze of humans to facilitate communication and collaboration.
- Gesture: As humans do, robots can use gestures to convey information and emotions. There are different types of gestures, such as:
  - *Deitic gestures*: Gestures that refer to a specific object or location in the environment, such as pointing.
  - *Iconic gestures*: Gestures that are used along with speech to convey information, such as using hand movements to illustrate a concept.
  - *Symbolic gestures*: Gestures that have a specific meaning within a culture, such as a thumbs-up.
  - *Beat gestures*: Gestures that are used related to the rhythm of speech.
- Touch: Even though touch is a powerful form of nonverbal communication, it is also the most difficult one to implement in robots as it means invading the personal space of humans. In some cases, they are however useful:
  - When the robot is zoomorphic, as it can be designed to be touched in a way that is similar to how humans interact with animals.
  - In nursing and caregiving robots, as touch can be used to provide comfort and support to patients.
- Posture and movement: The posture of the robot can also be used to convey information and emotions.

2.5. Human-Robot Teams

In some cases, humans and robots may need to work together as a team to achieve a common goal. Team members have the following characteristics:

- Share one or more common goals.
- Independently perform tasks that contribute to the common goal.
- Interact socially to coordinate their efforts and achieve the common goal.

Therefore, robots are also expected to be good team members. In order to know if a robot is felt as an actuall team member, two different concepts should be taken into account:

- Team Identity: Shared identity between a group of individuals.
- Team Identification: The degree to which an individual identifies (feels a sense of belonging) with a team.

Team Identification is crucial for effective teamwork, as it can lead to increased motivation, cooperation, and performance. Therefore, robots should be designed to foster team identification among human team members. *Trust* is also a crucial aspect of teamwork, so robots should be designed to build and maintain trust with human team members. There are different types of questionnaires that can be used to evaluate trust in the robot.

3. Human AI Interaction

AI systems are becoming increasingly prevalent in our daily lives, from virtual assistants to recommendation systems. As these systems become more sophisticated, it is crucial to understand how humans interact with them and how to design interfaces that facilitate effective communication and collaboration between humans and AI.

3.1. What is Artificial Intelligence?

Artificial Intelligence (AI) refers to the simulation of human intelligence in machines that are programmed to think and learn. AI systems can perform tasks that typically require human intelligence, such as visual perception, speech recognition, decision-making, and language translation. Even though intelligence in humans can be *measured*, AI can't be measured. There are however various tests that have been proposed to evaluate if a machine can be considered intelligent. The importance of these tests is shown in the Chinese Room thought experiment.

- Chinese Room Experiment:

  The Chinese Room thought experiment involves a person who does not understand Chinese sitting in a room with a set of instructions for how to respond to Chinese characters. The person receives Chinese characters as input and uses the instructions to produce appropriate responses, which are then sent back out of the room.

  From the outside, it may appear that the person in the room understands Chinese, but in reality, they are simply following a set of rules without any comprehension of the language. This thought experiment raises questions about whether a machine can truly understand language or if it is simply following a set of instructions.

The most well-known test for evaluating machine intelligence is the Turing Test, proposed by Alan Turing:

- Turing Test:

  In the Turing Test, a human evaluator interacts with both a machine and a human without knowing which is which. If the evaluator cannot reliably distinguish between the machine and the human, then the machine is said to have passed the Turing Test and is considered to be intelligent.

  It only tests for the ability to mimic human behavior, not for true understanding or consciousness.

3.2. Human-AI Interaction

Given the increasing prevalence of AI systems in our daily lives, it is important to understand how humans interact with these systems. Human-AI interaction refers to the ways in which humans and AI systems communicate and collaborate to achieve a common goal. There are three main aspects that need to be considered:

- Ethically Aligned Design: AI systems should be designed to align with human values and ethical principles, ensuring that they are used in ways that benefit society as a whole.

- Human Factors Design: AI systems should be designed with human factors in mind, taking into account the cognitive, social, and emotional needs of users to ensure that they are easy to use and understand. This is where *Explainable AI* comes into play, as we will later discuss.

- Technology Enhancement Design: AI systems should be designed to enhance human capabilities and performance, rather than replace them.

3.3. User's control

When designing AI systems, it is important to consider the user's control over the system. AI's goal is usually translating the raw data into a model that can be used to make predictions or decisions. However, there are different levels of control that users can have over this process:

- An algorithm that is fully automated does this conversion without any user input, so no human control is involved.

- Humans provide feedback to the training algorithm, which can be used to improve the model's performance. This is often referred to as *human-in-the-loop* learning.

- Users replace the training data or algorithms with their own knowledge and expertise, effectively taking all the control of the learning process, but having to do all the work.

Some researches argue that there is a *trade-off between the level of control and the automation of the system*. Fully automated systems may be more efficient and require less effort from users, but they may also be less transparent and less customizable. On the other hand, systems that allow for more user control may be more flexible and adaptable to individual needs, but they may also require more effort and expertise from users. However, some researchers argue that this trade-off is not necessarily inherent in the Human-AI interaction. This idea is shown in Figure 3.1, where the diagonal line represents the trade-off between control and automation, but there are also areas where both can be achieved simultaneously.

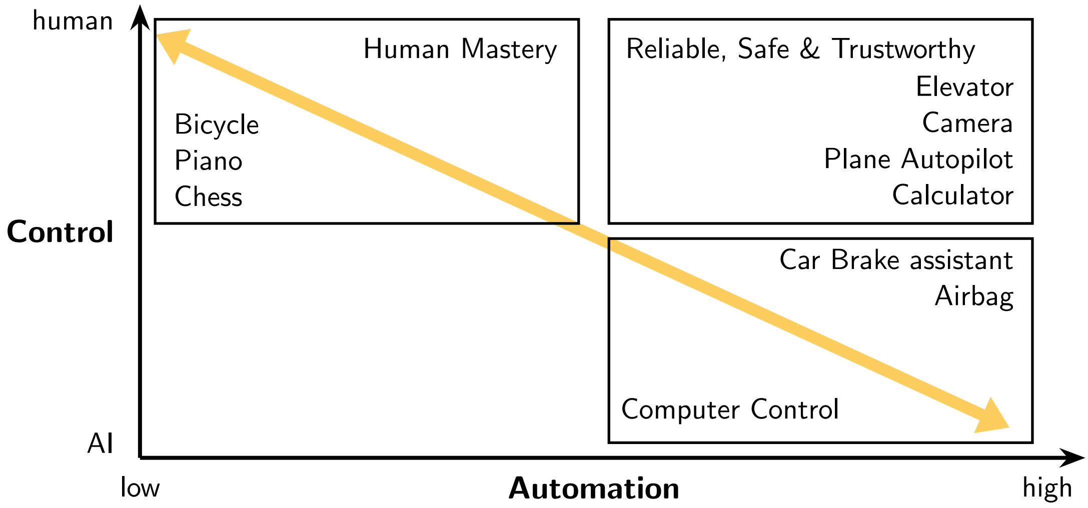


Figura 3.1: Control vs Automation in Human-AI Interaction

3.4. Explainability in AI Systems

Explainability refers to the ability of an AI system to provide clear and understandable explanations for its decisions and actions. There are several reasons why explainability is important in AI systems:

- Trust: Users are more likely to trust an AI system if they can understand how it works and why it makes certain decisions.
- Accountability: Explainable AI systems allow decisions to be audited and held accountable, which is especially important in high-stakes applications such as healthcare and finance.
- Ethical Considerations: Explainability can help ensure that AI systems are used ethically and responsibly, as it allows users to understand the potential consequences of the system's actions.
- Debugging and Improvement: Explainable AI systems can provide insights into how the system is functioning, which can be useful for debugging and improving the system over time.

There are different recipients of explanations in AI systems, including:

- Users affected by the model's decisions: These users may require explanations to understand how the model's decisions impact them and to ensure that they are treated fairly.
- Data scientists and developers: These individuals may require explanations to understand how the model is functioning and to identify potential issues or areas for improvement.
- Regulators and policymakers: These stakeholders may require explanations to ensure that AI systems are compliant with regulations and ethical standards.

- Domain experts or users of the system: These users may require explanations to understand how the AI system is making decisions in their specific domain and to ensure that it is providing accurate and relevant information.

- Managers and decision-makers: These individuals may require explanations to understand the implications of the AI system's decisions and to make informed decisions based on the system's outputs.

There is however an important trade-off between the accuracy of an AI system and its explainability. Highly accurate AI systems may be less transparent and harder for users to understand, which can lead to a lack of trust. On the other hand, more explainable AI systems may be less accurate but can foster greater trust among users. This is the goal of the field of Explainable AI (XAI), which aims to develop AI systems that are both accurate and explainable. Depeding on the application and the users, there are different types of explanations that may be more appropriate, such as:

- Analytical Explanations: These explanations provide a detailed analysis of the model's decision-making process, including the features and data that influenced the decision.

- Visual Explanations: These explanations use visualizations to help users understand the model's decision-making process, such as heatmaps or feature importance graphs.

- Rule-Based Explanations: These explanations provide a set of rules or guidelines that the model follows to make decisions, which can be easier for users to understand. It is presented as a simple `if-then` statement. It may be too simplistic for complex models, but it can be useful for certain applications where interpretability is more important than accuracy.

- Textual Explanations: These explanations provide a natural language description of the model's decision-making process, which can be more accessible for users who are not familiar with technical details. There are different types of textual explanations, such as:

  - Contrastive Explanations: These explanations focus on why a particular decision was made instead of another, highlighting the differences between the two outcomes.
  - Counterfactual Explanations: These explanations focus on what changes would need to be made to the input data to produce a different outcome, providing insights into the model's decision boundaries.

There are four steps that should be followed to design effective explanations for AI systems:

- Identify the target audience: Understanding who will be receiving the explanations is crucial for designing explanations that are appropriate and effective for that audience.

- Determine the purpose of the explanation: The purpose of the explanation will guide the content and format of the explanation, whether it is to build trust, provide accountability, or facilitate debugging and improvement.
- Choose the content of the explanation: The content of the explanation should be relevant and informative for the target audience, providing insights into the model's decision-making process without overwhelming users with technical details.
- Select the format of the explanation: The format of the explanation should be accessible and understandable for the target audience, whether it is through visualizations, textual descriptions, or rule-based explanations.

A last important consideration when designing explanations for AI systems is reliance on the AI's answers.

- Over-reliance: Users may become overly reliant on the AI system's outputs, leading to a lack of critical thinking and potential errors if the system provides incorrect information.
- Under-reliance: Conversely, users may under-rely on the AI system's outputs, leading to missed opportunities for improved decision-making and efficiency.

Appropriate explanations can help mitigate these issues by providing users with the necessary information to make informed decisions about when to trust the AI system and when to exercise caution.

4. Augmented Reality

Augmented Reality (AR) is a technology that superimposes computer-generated content onto the real world, enhancing the user's perception and interaction with their environment. The general term for this technology is Mixed Reality (MR), and its different types are defined in the *Reality-Virtuality Continuum*. The continuum describes the spectrum of environments that can be created by combining real and virtual elements, ranging from the completely real environment to the completely virtual environment. The different levels of the continuum are defined as follows:

- Real Environment: The physical world that we live in, which can be seen and interacted with directly.
- Augmented Reality (AR): The virtual content is overlaid onto the real world, allowing users to see and interact with both the real and virtual elements simultaneously.
- Augmented Virtuality (AV): The real world is integrated into a virtual environment, allowing users to interact with real-world objects within a virtual space.
- Virtual Environment: A computer-generated environment that may (or may not) resemble the real world, and is typically experienced through a device such as a headset or smartphone.

4.1. Devices for Augmented Reality

In order to experience Augmented Reality, users typically require specific devices that can capture the real world and display the augmented content. Some common devices for AR include Smartphones, Tablets, AR Glasses, and Head-Mounted Displays (HMDs). Each of these devices has its own advantages and limitations in terms of portability, field of view, and interaction capabilities. Some aspects to consider when choosing an AR device include:

Video vs Optical See-Through

- Video see-through (VST): devices use cameras to capture the real world, and then computer-generated content is superimposed onto the video feed.

  It can be implemented using a smartphone or tablet, where the device's camera captures the real world and the screen displays the augmented content. It also allows a full peripheral view of the real world, which may be needed in dangerous situations.

- Optical see-through (OST): devices have two layers: a transparent layer that allows users to see the real world directly, and a reflective layer that projects virtual content onto the transparent layer.

  OST devices typically require specialized hardware such as AR glasses or HMDs, which can be more expensive and less portable than VST devices. It can also provide a more immersive experience, but may suffer from latency and lower image quality compared to VST. Lastly, users may suffer from motion sickness or eye strain when using OST devices for extended periods of time.

HMD vs Mobile vs Projector

There are different specific devices for AR, each with its own advantages and disadvantages. Each one has a spatial relationship to the user and the real environment, which needs parameters to be known. There are two types of parameters:

- C: **C**onstant and **C**alibrated parameters that remain stable during usage.
- T: **T**racked parameters that change constantly and therefore need tracking.

Depending on the device, the parameters that need to be tracked may differ.

- HMDs used for OST: The distance to the real world needs to be tracked (T), while the distance to the eyes is constant (C). Depending if it uses eye-tracking or not, that mey be tracked (T) or constant (C).
- HMDs used for VST: The distance to the real world needs to be tracked (T), and the distance between the camera and the display is constant (C), as they are integrated. The distance to the eyes is either constant (C) or tracked (T), depending on whether the device uses eye-tracking or not.
- Mobile devices: The distance to the real world needs to be tracked (T), while the distance from the camera to the device is constant (C), as they are integrated. The distance to the eyes usually does not matter.
- Projectors: The eyes do not matter, as the content is projected onto a surface. The distance to the real world is either constant (C) or tracked (T), depending on whether the scene is static or dynamic.

Occularity in HMDs

- Monocular: The virtual content is displayed to only one eye, which can lead to a less immersive experience but is cheaper and less complex to implement.
- Bi-ocular: The virtual content is displayed to both eyes, but the same image is shown to each eye. This can provide a more immersive experience than monocular devices, but may still lack depth perception.

- Binocular: The virtual content is displayed to both eyes, with each eye receiving a slightly different image to create a sense of depth and immersion. This can provide the most immersive experience, but is also the most complex and expensive to implement.

Distance to the display

- Head Space: The display is located close to the user's head, such as in HMDs.
- Body Space: The display is located at a distance from the user's body, such as in mobile or handheld devices.
- World Space: The display is located in the environment, such as stationary displays or projectors.

Field of View (FOV)

The FOV is the extent of the observable world that can be seen at any given moment. A wider FOV can provide a more immersive experience, but may also require more processing power and higher-quality hardware to maintain image quality and reduce latency. Humans real FOV is usually larger than most AR devices can provide, which can lead to a less immersive experience and may cause users to feel disconnected from the augmented content.

4.2. Making sense of the real environment

In order to effectively augment the real environment, AR systems need to be able to understand and interpret the physical world. This involves several key components, as registration, tracking, and calibration.

4.2.1. Registration

Registration is the process of aligning the virtual content with the real world in a way that appears seamless and natural to the user. In order to do so, the real-world coordinate systems need to be mapped to the virtual coordinate systems. Several transformations are needed to achieve this mapping, including:

- Model Transformation: This transformation maps the virtual content from its own coordinate system to a common world coordinate system.

  If the real objects move and the virtual content needs to be registered to them, tracking is needed to update the model transformation in real time.

- View Transformation: This transformation maps the world coordinate system to the user's viewpoint, taking into account the user's position and orientation in the real world.

  If the camera is moving, tracking is needed to update the view transformation in real time.

- Projection Transformation: This transformation maps the 3D virtual content onto the 2D display of the AR device, taking into account the device's field of view and other display parameters.

  Calibration is needed to determine the parameters of the projection transformation.

Depending on where the objects are registered, there are different types of registration:

- Object Registration: The virtual content is registered to specific objects in the real world, such as a table or a building. If the objects are moving, tracking is needed to update the registration in real time.
- View Registration: The virtual content is registered to the user's viewpoint, so that it appears to be anchored in the real world regardless of the user's position or orientation. If the camera is moving, tracking is needed to update the registration in real time.
- World Registration: The virtual content is registered to a fixed location in the real world, such as a specific point on the ground or a landmark. Regardless of whether the user or the objects are moving, no tracking is needed to maintain the registration, as it is anchored to a fixed location in the real world.

In addition, there are two main types of registration techniques, depending on how well objects are registered:

- Geometric Registration: it focuses on just placing the virtual content in the correct position and orientation in the real world.
- Photometric Registration: it focuses on matching the lighting and shading of the virtual content to the real world, in order to create a more seamless and realistic integration.

  It is much more computationally expensive than geometric registration, as it requires real-time analysis of the lighting conditions in the real world and the ability to adjust the virtual content accordingly.

Lastly, *occlusion* should be considered in registration, as it can affect the realism of the augmented content. If virtual objects are not properly occluded by real objects, it can break the illusion of integration and make the augmented content appear less believable. Proper occlusion handling requires accurate depth sensing and tracking of both the real and virtual objects in the environment.

4.2.2. Tracking

Tracking is the process of continuously monitoring and measuring the real-world environment and the user's position and orientation. There are different types of tracking systems, including:

- Stationary Tracking: The tracking system is fixed in a specific location and tracks the user's position and orientation relative to that location.

- Mobile Tracking: The tracking system is integrated into the AR device and tracks the user's position and orientation relative to the real world.

- Optical Tracking: The tracking system uses cameras and computer vision algorithms to track the user's position and orientation based on visual markers or features in the real world.

There are two main approaches to tracking in AR:

- Model-Based Tracking: This approach uses a pre-defined model of the real world, such as a 3D map or a set of known landmarks, to track the user's position and orientation. The tracking system compares the real-world environment to the model and updates the user's position and orientation accordingly.

- Model-Free Tracking: This approach does not rely on a pre-defined model of the real world, but instead uses real-time sensor data to track the user's position and orientation.

Different specific tracking techniques include:

- Marker-Based Tracking: This technique uses visual markers, such as QR codes or fiducial markers, placed in the real world.

- Natural Feature Tracking: This technique uses computer vision algorithms to track natural features in the real world, such as edges, corners, or textures.

  An specific example of this technique is Simultaneous Localization and Mapping (SLAM), which combines:

  - Visual natural attribute tracking
  - Depth sensing using specialized sensors.

One specific object that usually needs to be correctly tracked is the user's hands and fingers, as they are often used for interaction with the virtual content.

4.2.3. Calibration

Calibration is the process of determining the parameters of the AR system, such as the position and orientation of the cameras, the field of view, and the display characteristics. Proper calibration is essential for accurate registration and tracking, as it ensures that the virtual content is correctly aligned with the real world and that the user's movements are accurately tracked.

4.3. Situated AR

An important aspect of AR is the concept of *situated AR*, opposite to conventional AR.

- <u>Conventional Visualization</u>: The virtual content is displayed without integration with the real world. This includes traditional displays, or even AR content that is simply overlaid on the real world.

  This includes context registered relatively to markers, as they are added to the real world but do not have a specific meaning in it.

- <u>Situated Visualization</u>: The virtual content is integrated into the real world in a way that is meaningful and relevant to the user's context.

Some situated AR techniques include:

- <u>Annotations and Labels</u>: Virtual content is used to provide additional information about real-world objects or locations, such as labels, descriptions, or instructions.

  They should not cover real-world objects nor overlap with other annotations.

- <u>X-Ray Visualization</u>:

  This technique allows users to see through real-world objects, providing a view of hidden or obscured content. Transparency should still be used to keep up occlusion and depth impression.

  Some specific examples of x-ray visualization include showing the internal components of a machine, or allowing users to see through walls to detect wires or pipes.

- <u>Spatial Manipulation</u>:

  This technique allows users to manipulate virtual content in the real world, such as moving, resizing, or rotating virtual objects. It can be used for tasks such as interior design, where users can visualize how furniture would look in a room before making a purchase.

- <u>Information Filtering</u>:

  This technique allows users to filter or hide certain real-world objects or information, in order to focus on specific aspects of the environment. There are two main types of information filtering:

  - <u>Knowledge-Based Filtering</u>: This technique uses the user's knowledge and preferences to filter information.

    For instance, combined with X-Ray Visualization, it can be used to show only specific types of information, such as showing only the water pipes in a building while hiding the electrical wires.

  - <u>Spatial Filtering</u>: Use geometric and geographic information to place information. For instance, it can be used to show information about nearby points of interest while hiding information about distant locations.

4.4. AR Interaction

Interaction in AR can be achieved through various input methods and output modalities.

4.4.1. Output Modelities

There are three main output modalities in AR:

- Head-Referenced Display: The virtual content is displayed in a way that is anchored to the user's FoV, such as in smart glasses.
- Hand-Referenced Display: The virtual content is displayed in the user's palm of hand(s), allowing for direct interaction with the augmented content.
- Agile Display: The virtual content is displayed everywhere and with changing perspectives of users, such as in mobile displays (carried around or worn) or stationary displays (adaptive projectors).

Avatars

Avatars are virtual representations of users in the AR environment, allowing for social interaction and communication with other users. Some different aspects that should be taken into account when designing avatars include:

- Full or half body representation.
- Full featured or simplified abstract representation.

  Even though a full featured avatar can provide a more immersive experience and are usually more trusted, they can also be more expensive and complex to implement, and may require more processing power and higher-quality hardware to maintain image quality and reduce latency.
- Life size or scaled representation.

  Miniature avatars are usually preferred over life-size ones, and a potential reason for this is that they can fit in the user's FoV more easily.

4.4.2. Input Modalities

There are several input modalities for AR interaction, including:

- Rigid Object Tracking: This technique tracks the position and orientation of your head (HMD) or different devices (mobile, projectors) to allow for interaction with the virtual content.
- Gesture Recognition: Using hand tracking, the system can recognize specific hand gestures to trigger interactions with the virtual content. This gesture recognition should be as intuitive and natural as possible.
- Pointing: A laser beam or a ray can be emitted from the user's hand or device to point at the AR scene, allowing for selection of objects and non-verbal communication.
- Drawing: Similar to pointing, but instead of just selecting objects, the user can draw in the air to create virtual content or annotate the real world.
- Touch: As in real interfaces, virtual buttons or controls can be placed in the AR scene, allowing users to interact with them through touch.

The Midas Touch Problem

An important issue in AR interaction is selecting objects in the AR scene.

- Selection via Gaze

  A first approach to selection was using the user's gaze, where the system tracked the user's eye movements to determine where they were looking in the AR scene.

  However, this lead to the *Midas*[1] *Touch Problem*, where every object that the user looked at would be selected, making it difficult to interact with the AR content without accidentally selecting things.

- Selection via Pointing or Gestures

  In order to solve the Midas Touch Problem, selection can be done through pointing or gestures, where the user explicitly indicates which object they want to interact with.

[1]King Midas was a figure in Greek mythology who was granted the ability to turn everything he touched into gold. Even though this power was initially seen as a blessing, it quickly became a curse, as Midas could not eat, drink, or interact with anything without turning it into gold.

5. Designing Interactive Systems: Methods

In order to design effective interactive systems, the human should always be at the center of the design process. The specific phases of the design process are the following:

1. Understand and Specify the Context of Use

   Understanding the context of use is a critical step in the design process. It involves identifying the users, their tasks, and the environment in which the system will be used. This phase helps designers to gather relevant information about user needs, preferences, and limitations, which will inform the design decisions.

2. Specify the User Requirements

   Once the context of use is understood, the next step is to specify the user requirements from their perspective. This involves defining what the users need from the system. User requirements should be clear, measurable, and prioritized to guide the design process effectively.

3. Design Solutions that Meet User Requirements

   In this phase, designers create solutions that address the specified user requirements. Three stages are involved in this process, but the process may go back and forth between these stages:

   a) Early Design: storyboards or use cases can be used to illustrate the design concepts and gather feedback from users.

   b) First Drafts: designers create initial low-fidelity prototypes or mockups of the system.

   c) Refined Designs: designers develop high-fidelity prototypes that closely resemble the final product, incorporating feedback from users and stakeholders to refine the design.

4. Evaluate the Design Against User Requirements

   The final phase involves evaluating the design to ensure that it meets the user requirements. This can be done through usability testing, where the system is tested with real users to identify any issues or areas for improvement. These people are asked to "think aloud" while using the system, and their feedback is used to make necessary adjustments to the design.

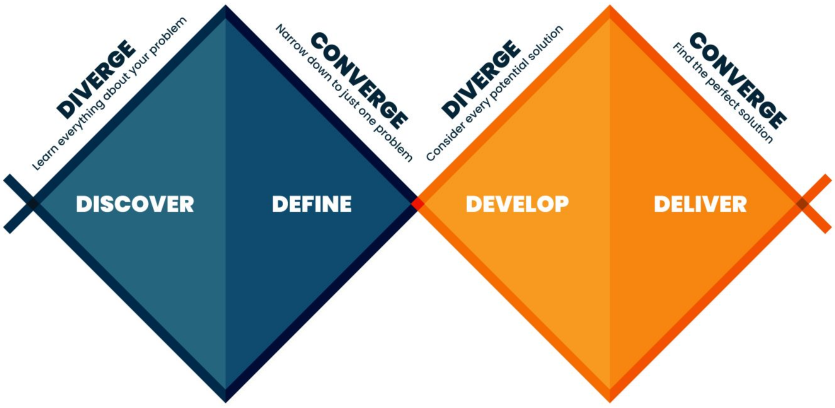


Figura 5.1: The Double Diamond model of the design process.

The Double Diamond model, shown in Figure 5.1, is a visual representation of the design process. It represents the divergent and convergent thinking that occurs during the design process, with two diamonds representing the phases of understanding and defining the problem, and developing and delivering the solution.

5.1. Participation in the Design Process

The participation of the stakeholders in the design process is crucial for creating effective interactive systems. Some aspects of participation include:

- Duration and frequency of their participation
- Influence on the design decisions
- Direct involvement in the design activities or indirect involvement through representatives
- The level of expertise and knowledge of the stakeholders

The choice of stakeholders is crucial for the success of the design process. It is important to involve a diverse group of stakeholders with different perspectives and expertise to ensure that the design meets the needs of all users.

Personas are fictional characters that represent different user types within a targeted demographic. They are created based on user research and help designers understand the needs, goals, and behaviors of their users. By using personas, designers can make informed decisions about the design of the system and ensure that it meets the needs of the intended users.

5.2. Prototyping

Prototyping is a crucial step in the design process that allows designers to create a preliminary version of the system. Prototypes are systems that are not fully functional but allow designers to test and evaluate the design concepts. Different types of prototypes can be used:

- Storyboard: a visual representation of the user interface and interactions, often used to illustrate the flow of the system.
- Mockup: a simple prototype that represents the visual design of the system, but does not have any functionality.
- Horizontal Prototype: a prototype that represents the overall structure and layout of the system.
- Vertical Prototype: a prototype that represents a specific feature or functionality of the system in detail.
- Low-Fidelity Prototype: a prototype that is simple and inexpensive to create, often made with paper or cardboard.
- Paper prototype: a low-fidelity prototype made with paper or cardboard that allows designers to test the layout and functionality of the system.
- High-Fidelity Prototype: a prototype that closely resembles the final product, with more advanced functionality and interactivity.
- Wizard of Oz Prototype: a prototype that simulates the functionality of the system, but is actually controlled by a human operator behind the scenes.

5.3. Design Thinking

Design thinking is a human-centered approach to problem-solving that is based on seeing the problem from the user's perspective. It has six phases, which are:

1. <u>Understand</u> the problem within the team.
2. <u>Observe</u> potential users and gathering insights about their needs and behaviors.
3. <u>Synthesize</u> the results of the previous phases. Personas and scenarios can be used to represent the users and their needs.
4. <u>Ideation</u> to generate a wide range of ideas and potential solutions to the problem. Brainstorming and other creative techniques can be used to encourage divergent thinking, and no "but" or "if" statements are allowed during this phase. No limits to crazy ideas are allowed.
5. <u>Prototyping/testing</u> potential solutions to gather feedback and refine the design.
6. <u>Implementation</u> of the final solution, which involves developing and deploying the system to users.

6. Theories and Frameworks for Design

When designing interactive systems, it is important to have a solid understanding of the theories and frameworks that underpin the design process. These theories provide a foundation for understanding user behavior, cognitive processes, and the principles of effective interaction design. In this chapter, we will explore several key theories and frameworks that are commonly used in the field of interactive systems design.

6.1. Activity Theory

The Activity Theory model provides a comprehensive framework for understanding human activities in context. It provides three different hierarchical levels of acting:

1. Activity: This is the highest level of analysis and refers to the overall *motive* that drives a person to engage in a particular action. Activities are typically long-term.

2. Action: This level focuses on the *goal-directed* processes that individuals undertake to achieve specific goals. Actions are more immediate and are often part of a larger activity.

3. Operation: This is the most granular level of analysis and refers to the *automatic* or habitual processes that individuals perform as part of an action. Operations are often unconscious.

The Activity Theory model is represented in Figure 6.1. It consists of several key components:

- Subject: The individual or group engaged in the activity. It interacts with the object through the use of instruments according to the rules and within the context of the community.

- Object: The target and focus of the subject. It isn't neccessarily a physical object (e.g., questions) and its state usually changes as a result of the activity, so an outcome is produced.

- Instruments: The tools or mediating artifacts that the subject uses to interact with and change the object. Instruments can be physical tools, symbols, or concepts.

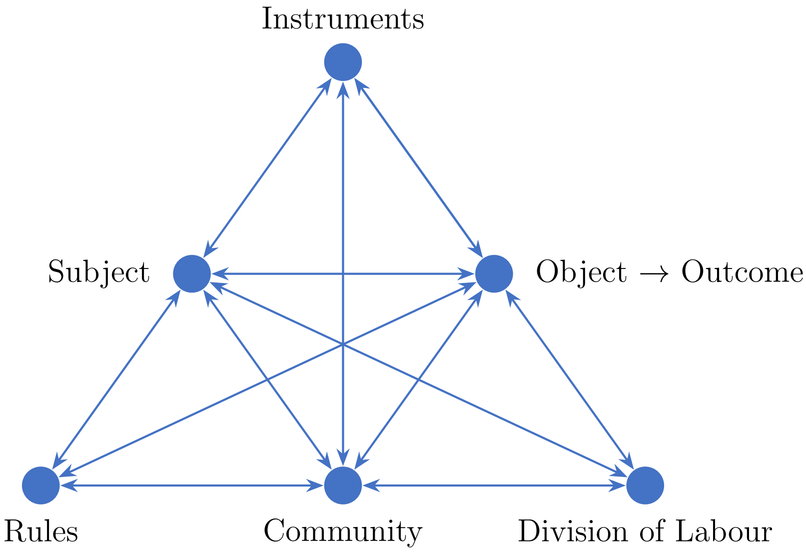


Figura 6.1: Activity Theory Model

Depending on the context, an instrument can be an object in itself (as in the case of a software application, when the aim is creating it instead of using it).
Rules: The norms, conventions, and regulations that govern the activity.

- Community: The group of individuals who share the same object and are involved in the activity.

- Division of Labour: The distribution of tasks and responsibilities among members of the community.

6.2. Attribution Theory

Attribution Theory is a psychological framework that seeks to explain how individuals interpret and assign causes to events and behaviors. It is particularly relevant in the context of interactive systems design, as it helps designers understand how users perceive and respond to system feedback, errors, and successes. It classifies attributions depending on different dimensions:

Internal vs. External This dimension distinguishes between causes that are attributed to the individual's own characteristics (internal) and those that are attributed to external factors, such as the environment or other people.

Stable vs. Variable This dimension differentiates between causes that are perceived as stable and unchanging over time (stable) and those that are seen as temporary or fluctuating (variable).

Controllable vs. Uncontrollable This dimension assesses whether the individual percieves the cause of an event as something they can influence or control (controllable) or as something beyond their control (uncontrollable).

In addition, the theory also considers the role of covariation in attributions. Covariation refers to the relationship between an effect and its potential causes over time. Three aspects are considered when analyzing covariation:

- Consistency: This aspect examines whether the attribution is similar to prior attributions made in similar situations. High consistency suggests that the cause is likely to be stable and reliable.

- Consensus: This aspect looks at whether the attribution is similar to attributions made by others in similar situations. High consensus indicates that the cause is widely recognized and accepted.

- Distinctness: This aspect assesses whether the attribution is directly associated with the specific situation. High distinctness suggests that the cause is unique to the particular context.

This theory is useful for designers to understand how users interpret system behavior and how they attribute success or failure to their own actions or to the system itself. If errors occure, describing them will help avoiding internal attributions, which can lead to frustration and negative user experiences.

6.3. Motivation Theory

Motivation Theory explores the factors that drive individuals to engage in certain behaviors and activities. In the context of interactive systems design, understanding user motivation is crucial for creating engaging and effective experiences. Two different approaches to motivation are commonly discussed: intrinsic and extrinsic motivation.

- Intrinsic Motivation: This type of motivation arises from within the individual and is driven by personal interest, enjoyment, or a sense of accomplishment. Users who are intrinsically motivated are more likely to engage with a system for the sheer pleasure of using it, rather than for external rewards or incentives.

  The *Self-Determination Theory* (SDT) is a widely recognized framework that explains intrinsic motivation. According to SDT, individuals have three basic psychological needs that, when satisfied, enhance intrinsic motivation:

  1. Competence: The need to feel effective and capable in one's actions. When users perceive that they are mastering a task or skill, their intrinsic motivation is enhanced.
  2. Autonomy: The need to feel a sense of control and choice over one's actions. When users have the freedom to make decisions and act according to their own preferences, their intrinsic motivation is strengthened.
  3. Relatedness: The need to feel connected and supported by others. When users experience a sense of belonging and social connection, their intrinsic motivation is enhanced.

- Extrinsic Motivation: This type of motivation is driven by external factors, such as rewards, recognition, or social pressure. Users who are extrinsically motivated may engage with a system to achieve specific outcomes or to gain approval from others, rather than for personal satisfaction.

6.4. Place and Space Theory

When designing interactive systems, it is important to consider the concepts of place and space, as they influence how users perceive and interact with their environment. An important distinction is made between the two concepts:

- Space: This refers to the physical environment that might enable or constrain user interactions. It encompasses the layout, dimensions, and spatial relationships of the environment in which users operate.

- Place: This refers to the social and cultural context in which interactions occur. The space acquire meaning and significance through the activities, experiences, and social interactions that take place within it, resulting in a sense of place.

That way, the space is the opportunity for interaction, while the place is the context in which the interaction occurs. Designers must consider both aspects in order to achieve their created space to be perceived as a meaningful place by the users.

7. Exercises

7.1. Research Methods for Interactive Systems Design

Ejercicio 7.1.1. Briefly describe how ethnography works. Highlight its advantages and limitations, and give an example application in a real-world scenario where it would be particularly useful.

Ejercicio 7.1.2. Compare the roles of the researcher in Ethnography, Action Research, and Research through Design. What are the similarities and differences?

Ejercicio 7.1.3. State the advantages and disadvantages of two of the research methods that were presented in the lecture. In this context, explain why combining methods might be beneficial. What is this process called?

Ejercicio 7.1.4. Imagine you are developing a platform for interaction between different people. The platform concept is ready and you want to gain user feedback. Which research method(s) do you use and why?

Ejercicio 7.1.5. A new social robot is developed. It is supposed to help seniors in an elderly home and also support them with being more active in their daily life. Which research methods should be used to evaluate the acceptance of the robot and its impact on the elderly? Describe the research design.

Ejercicio 7.1.6. You are an employee in a company, and your boss has assigned you the task of gathering research data for a new program that will be used in schools. Select one of the research methods and explain why this method is appropriate for this scenario.

Ejercicio 7.1.7. You want to gather knowledge about how people in an office communicate with each other and distribute different tasks between each worker. Later on, you want to design a new technical communication tool to make communication and task distribution easier. Which research method would you use to gather insights and why?

Ejercicio 7.1.8. Please indicate for the following cases below which study method you would try to use to solvethe following issue and why, as well as how many participants (approximately) you would recruit and why. Consider everything you have learned in the lecture and teaching cases.

1. You want to design a new HoloLens application that is supposed to help care workers during their daily activities in intensive care units. However, your knowledge about the domain and working conditions is very limited. So, you decide to gather some knowledge about intensive care units first. How would you want to gather knowledge to design your application?

7.2. Human Robot Interaction

Ejercicio 7.2.1. What is the difference between coexistence, cooperation, and collaboration in HRI (according to Schmidtler et al., 2015)? Give one example task for each.

Ejercicio 7.2.2. Explain the Sense-Think-Act paradigm in robotics. How does each component contribute to the overall autonomy of a robot, and what would happen if one component fails in a human-robot interaction scenario?

Ejercicio 7.2.3. Please read through the scenario and discuss whether the interactive system is a robot or not according to the Sense-Think-Act definition.

Ejercicio 7.2.4. In order for humans to really collaborate with robots, the robots have to communicate their intents to the humans. Describe two ways in which a robot could communicate the intent to give an object to a human and discuss the pros and cons.

Ejercicio 7.2.5. Compare verbal and nonverbal interaction in HRI. Give two examples of nonverbal interaction methods.

Ejercicio 7.2.6. Why shouldn't we neglect nonverbal communication when designing a robot? Provide 2 reasons and examples for them.

Ejercicio 7.2.7. A robot is deployed in a classroom to invigilate students writing exams. How could the robot look? Give reasons for your answer.

Ejercicio 7.2.8. For a holiday job, students need to be taught how to correctly place goods in a supermarket. Instead of a human assistant, you want to use a robot to demonstrate and explain the tasks that need to be completed, as well as to check whether the trainee is missing any steps. What kind of robot would you use in this setting?

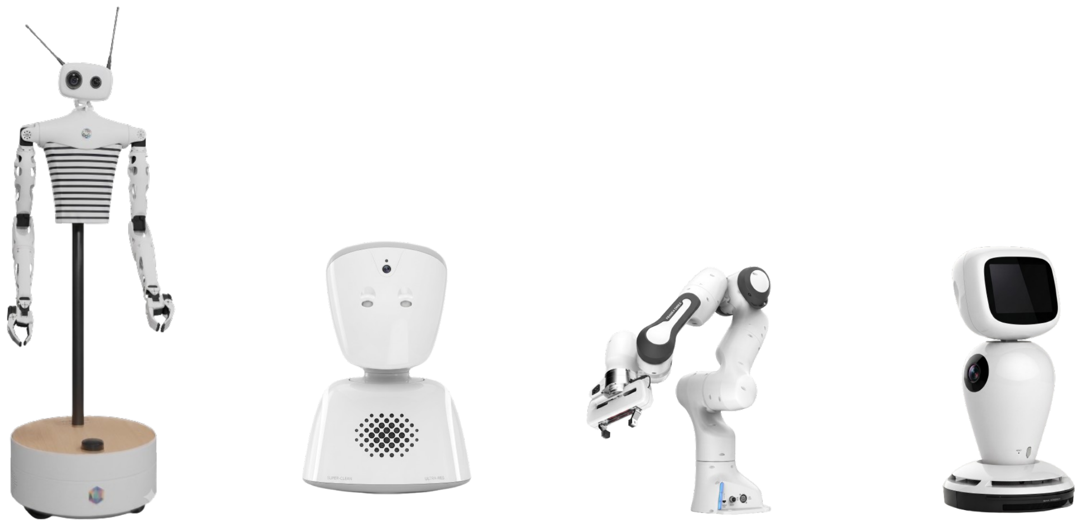
Ejercicio 7.2.9. How does the concept of "affordance" influence children's perception of educational robots, and how should it be considered in the design of such robots (e.g., telepresence robots for remote learners)?

Ejercicio 7.2.10. You are designing a robot to assist elderly people at home. Using the concepts of affordance, anthropomorphism, and social zones, outline key design considerations.

Ejercicio 7.2.11. You want to develop a robot that can help kids in school to learn about mathematics. Which morphological structure would you use to make the robot the best option for working with kids and why?

Ejercicio 7.2.12. Explain the difference between the "inside-out" and "outside-in" approaches in HRI. In the following scenario, which approach would you choose and why?

Ejercicio 7.2.13. Given the following two graphical representations, explain which one represents the "Uncaney Valley" effect and why. What do you have to consider when designing a robot to avoid this effect?



(a) Option A (b) Option B (c) Option C (d) Option D

Figura 7.1: Comparison of the four robot models.

Ejercicio 7.2.14. Given the robots in Figure 7.1, which one would you choose for the following scenarios and why?

- The robot is supposed to be an avatar for children in class who are currently hospitalized and cannot attend class.

7.3. Human AI Interaction

Ejercicio 7.3.1. Explain the three aspects (ethically-aligned design, human factors design, and technology enhancement) of Human-AI Interaction. Identify which aspect is neglected in the given scenario and justify your answer.

Ejercicio 7.3.2. Name two forms of explanations and describe them. Why are explanations important?

Ejercicio 7.3.3. We came to the conclusion that AI should be explainable. What are the 4 steps to explainability?

Ejercicio 7.3.4. A doctor receives a diagnosis suggestion from an AI but doesn't follow it. How and why can explanation modalities be used to increase trust and understanding?

Ejercicio 7.3.5. Briefly describe the thought experiment "Chinese room". What can you deduce about the intelligence of programs from this experiment?

7.4. Augmented Reality

Ejercicio 7.4.1. Compare model-free and model-based tracking in AR. How do both approaches work, what are the differences, and for which real-world application would each approach be suitable?

Ejercicio 7.4.2. Explain the concepts of "markers" and "SLAM" in AR. How do they differ and which technique (model-based or model-free) do each of them use?

Ejercicio 7.4.3. Name one of the key differences between Video See-Through (VST) and Optical See-Through (OST). Explain two advantages and two disadvantages of each. Name one scenario for which each would fit best.

Ejercicio 7.4.4. Explain the concepts of tracking and registration in Augmented Reality, detailing their role in overlaying virtual content onto the real world. Differentiate between geometric and photometric registration, providing a brief example for each. Why is low latency crucial for a good AR experience?

Ejercicio 7.4.5. List and explain two input modalities that can be useful to interact in AR. Name one advantage and one disadvantage for each.

Ejercicio 7.4.6. Which input modality (Rigid Object Tracing, Gestures, Pointing and Drawing, Touch) would you implement for this scenario? Name two reasons.

Ejercicio 7.4.7. In the lecture, various interaction techniques for augmented reality were discussed, such as gesture-based interaction, voice commands, and gaze monitoring. Consider the following use case and explain which interaction method you would recommend and why:

- <u>Use Case:</u> You are designing an AR application for elderly users that helps them perform light exercises at home. The app needs to be intuitive, minimize physical strain, and avoid requiring fine motor skills.

Ejercicio 7.4.8. Define what situated visualization in AR means, contrast it with conventional visualization methods, and provide two different techniques used for situated visualization along with an example use case for each.

Ejercicio 7.4.9. Given these three scenarios and these three gestures, which gesture would you use for each scenario and why?

- Possible gestures: *Pen gestures, Device gestures, Eye gestures, Hand gestures*

Ejercicio 7.4.10. During the lecture, gestural interaction techniques for AR were discussed. Explain two advantages and two disadvantages of using gestures as an input modality in AR.

Ejercicio 7.4.11. Describe the concept of "Reality-Virtuality Continuum".

Ejercicio 7.4.12. Explain the "Midas Touch" problem in AR. How can it be mitigated? Provide one example of a real-world application where this problem could arise and how it could be addressed.

7.5. Designing Interactive Systems: Methods

Ejercicio 7.5.1. You have been tasked with designing a social robot for a healthcare facility. How would you identify and involve the relevant stakeholders in the design process? Name two methods that help to involve stakeholders and explain also how these methods could be applied.

Ejercicio 7.5.2. You are researching the use of AI in the classroom. A chatbot with generative AI is to be used in the classroom by teachers as well as students. In the case of the stakeholder analysis, create at least two personas. Which personas that would be? Which kind of stakeholders?

Ejercicio 7.5.3. Name three characteristics of prototype and explain two differences between a prototype and a mock-up.

Ejercicio 7.5.4. Briefly describe the Wizard of Oz method and why it is needed. Give an example of a scenario where this method could be applied.

7.6. Theories and Frameworks for Design

Ejercicio 7.6.1. Give an explanation of the concept of internal and external attribution according to the Attribution Theory and provide an example for each in the context of HCI.

Ejercicio 7.6.2. Draw an activity theory triangle for the action "writing the exam" regarding the activity "visiting the course Interactive Systems". Then analyse your solution and name two different actions and operations in the activity.

Ejercicio 7.6.3. Given the following Scenario, explain and categorize the attributions that you could make (Internal/External & Stabile/Variable):

- You take part in a team competition and end up losing.